

Exploratory Data Analysis (EDA) Report

Project Title: Diamond Price Analysis

Introduction

In this project, I performed **Exploratory Data Analysis (EDA)** on the Diamond Price dataset. The main aim of this project is to understand the dataset, clean the data, and analyze the factors that affect diamond prices. EDA helps in finding useful information from the dataset before building a machine learning model.

Objective

The objectives of this project are:

- To understand the Diamond Price dataset.
- To check the dataset structure and data types.
- To identify missing values and duplicate records.
- To detect and remove outliers.
- To calculate statistical measures.
- To visualize the data using different graphs.

Dataset Description

First, I imported the required libraries such as **Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn**. Then I loaded the **Diamond Price dataset** using Pandas. I used functions like `head()`, `tail()`, `shape`, `columns`, `info()`, and `describe()` to understand the dataset. These functions helped me know the number of rows, columns, data types, and summary statistics.

Data Cleaning

After loading the dataset, I checked for missing values using `isnull().sum()` and checked for duplicate records using `duplicated().sum()`. I detected outliers in the **price** column using the **Interquartile Range (IQR)** method. The outliers were removed, and the price values were clipped within the lower and upper limits to improve the quality of the dataset.

Statistical Analysis

I calculated the following statistical measures:

- Mean
- Median
- Mode
- Variance
- Standard Deviation
- Correlation Matrix

These measures helped me understand the distribution of the numerical features and the relationship between them.

Data Analysis

I analyzed the distribution of different diamond **cut** and **clarity** categories using `value_counts()`. I also studied the relationship between **carat** and **price** using correlation analysis and scatter plots. This helped me understand which factors have a greater impact on diamond prices.

Data Visualization

To understand the dataset better, I created the following graphs:

- **Bar Chart** – Shows the number of diamonds in each cut category.
- **Histogram** – Displays the distribution of diamond prices.
- **Scatter Plot** – Shows the relationship between carat and price.
- **Line Chart** – Displays the variation in diamond prices across the dataset.

These visualizations made it easier to identify trends and patterns in the data.

Observations

From the analysis, I observed the following:

- The dataset was loaded successfully.
- Missing values and duplicate records were checked.

- Outliers in the price column were detected and handled using the IQR method.
- Diamonds with higher **carat** generally had higher prices.
- Different cut and clarity categories showed different distributions.
- The graphs helped in understanding the relationship between diamond features and price.

Conclusion

This project helped me understand the importance of Exploratory Data Analysis in data science. I learned how to inspect the dataset, clean the data, detect outliers, calculate statistical measures, and create visualizations. The analysis gave me a better understanding of the factors affecting diamond prices. After completing the EDA process, the dataset became clean and ready for building machine learning models for price prediction.